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The Prognostic Value of TyG Index and Right Ventricular Function in HFpEF Patients With Chronic Kidney Disease

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Abstract

Background: Right ventricular (RV) dysfunction and insulin resistance (IR) are associated with poor outcomes in heart failure with preserved ejection fraction (HFpEF). We hypothesized that the triglyceride–glucose (TyG) index, as a surrogate for IR, and RV function parameters would independently predict adverse outcomes in HFpEF, with potentially enhanced prognostic value in patients with chronic kidney disease (CKD), where RV impairment is more prevalent.

Methods: We prospectively enrolled 395 HFpEF patients admitted between October 2023 and October 2024. Two–dimensional RV free–wall longitudinal strain (2D–RVFWLS), three-dimensional RV ejection fraction (3D–RVEF), and TyG index were measured at baseline. The composite endpoint was cardiovascular (CV) death or HF rehospitalization. Multivariable Cox proportional hazards and Fine–Gray competing risk models were applied to identify predictors of adverse outcomes. Model performance was assessed using the C–index, category–free net reclassification improvement (cfNRI), and integrated discrimination improvement (IDI).

Results: In the overall cohort, lower 2D–RVFWLS and 3D–RVEF, along with higher TyG index, were associated with worse outcomes. RV dysfunction was more frequent and pronounced in the CKD group (lower absolute 2D–RVFWLS [16.7% vs. 19.5%, $P<0.001$]). During a median follow-up of 18.3 months, 158 patients experienced adverse outcomes. In multivariable Cox models, lower estimated glomerular filtration rate (eGFR) and smaller mitral E/A were independently associated with adverse outcomes. Adding the TyG index improved the base model (optimism–corrected ΔC–index = +0.013; apparent ΔC = +0.036), while the combination of TyG and 2D–RVFWLS achieved the highest predictive accuracy and significantly enhanced reclassification (cfNRI +0.362, IDI +0.396, both $p < 0.001$), particularly in the CKD subgroup.

Conclusions: In HFpEF, the TyG index and 2D–RVFWLS provide independent prognostic value superior to 3D–RVEF. Their combination offers robust risk stratification, particularly for the vulnerable phenotype of HFpEF with CKD.

Trial registration: Retrospectively registered in the Chinese Clinical Trial Registry, registration number pending.

Abbreviations: cfNRI, category–free net reclassification improvement; CKD, chronic kidney disease; DCA, decision curve analysis; eGFR, estimated glomerular filtration rate; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; IDI, integrated discrimination improvement; IR, insulin resistance; RV, right ventricle; RVEF, right ventricular ejection fraction; RVFWLS, right ventricular free–wall longitudinal strain; TyG, triglyceride–glucose index.

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1 | Introduction

Heart failure with preserved ejection fraction (HFpEF) accounts for approximately 50% of heart failure hospitalizations, and chronic kidney disease (CKD) is frequently observed in HFpEF [1, 2]. This cardiorenal overlap confers heightened risk of cardiovascular (CV) death and HF rehospitalization, driven by shared pathways including systemic inflammation, volume dysregulation, and metabolic stress [3].

Right ventricular (RV) dysfunction is a powerful determinant of outcomes in HFpEF, yet its accurate quantification remains challenging due to complex geometry and load dependency of conventional metrics [4]. Although three-dimensional RV ejection fraction (3D–RVEF) offers volumetric precision, its clinical utility is limited by technical constraints—particularly in breathless or obese patients common in HFpEF [5]. In contrast, two–dimensional RV free–wall longitudinal strain (2D–RVFWLS) provides a reproducible, angle-independent measure of myocardial deformation, with emerging prognostic relevance across HF phenotypes [6].

Insulin resistance (IR), a hallmark of the HFpEF metabolic phenotype, may directly impair RV performance through multiple mechanisms. First, IR induces mitochondrial dysfunction and reduced fatty acid oxidation, compromising RV energetics under pressure overload [7]. Second, hyperinsulinemia promotes myocardial fibrosis via activation of profibrotic pathways, disproportionately affecting the thin-walled RV [8]. In CKD, these processes are amplified by uremic toxins, oxidative stress, and Renin–Angiotensin–Aldosterone System (RAAS) activation, accelerating RV remodeling and diastolic stiffening [3, 9]. Despite this mechanistic plausibility, the interplay between IR and RV deformation in HFpEF—particularly under cardiorenal stress—remains underexplored.

The triglyceride–glucose (TyG) index, a surrogate marker of IR, correlates with myocardial fibrosis and adverse remodeling in HFpEF [10]. Recent studies link elevated TyG to CV mortality [11], but none have integrated it with advanced RV functional assessment, nor examined its incremental value in CKD subgroups using competing-risk frameworks.

Despite this mechanistic plausibility, the interplay between IR and RV deformation in HFpEF—particularly under cardiorenal stress—remains underexplored. Therefore, we hypothesized that 2D–RVFWLS and the TyG index independently predict adverse outcomes in HFpEF, with enhanced prognostic yield when combined—particularly in patients with CKD. To test this, we prospectively evaluated: (1) differences in RV function (2D–RVFWLS vs. 3D–RVEF) between HFpEF patients with and without CKD; (2) their relative prognostic performance; and (3) whether adding the TyG index improves risk discrimination and reclassification beyond echocardiographic parameters alone. This integrated cardiorenal–metabolic approach aims to refine prognostic stratification in a high-risk HFpEF population.

2 | Methods

2.1 | Study Design and Population

We conducted a single-center, prospective study at Gansu Provincial Hospital. Consecutive adults hospitalized with HFpEF between October 2023 and October 2024 were screened ($n = 493$). HFpEF was diagnosed according to the 2021 european society of cardiology (ESC) guidelines and required: (1) left-ventricular ejection fraction (LVEF) $\geq$ 50%; (2) typical signs/symptoms of HF; (3) elevated NT–proBNP > 220 pg mL^{-1}; and (4) structural and/or functional cardiac abnormalities with an HFA–PEFF score $\geq$ 5. Exclusions: significant valvular disease, prior cardiac surgery, active cancer or infection, and poor-quality 2D or 3D echocardiographic images. After these exclusions, 395 patients were included in the final analysis (Figure 1).

The study protocol was approved by the ethics committee of gansu provincial hospital (Approval No. 2025–566), since all data were collected for clinical purposes and no invasive procedures were performed, no specific written consent form was necessary for this study. All patients provided consent for the use of their anonymized data for research purposes.

2.2 | Baseline Data Collection

Demographics, comorbidities, new york heart association (NYHA) class, haemodynamic parameters, and medication use were obtained from the hospital's electronic medical records. Fasting blood samples ($\geq$ 8 h fast) were collected on the morning after admission and analyzed for total cholesterol, triglycerides, high-density lipoprotein cholesterol (HDL–C), low–density lipoprotein cholesterol (LDL–C), fasting blood glucose (FBG), serum creatinine, and N–terminal pro–B–type natriuretic peptide (NT–proBNP) using standardized enzymatic assays. The estimated glomerular filtration rate (eGFR) was calculated using the CKD–EPI equation [12], CKD was defined as a sustained eGFR < 60 mL/min/1.73 m^2 for more than 3 months, confirmed by reviewing patients' previous medical records to distinguish from acute kidney injury. The triglyceride–glucose (TyG) index was computed as ln [TG (mg/dL) × FBG (mg/dL) / 2] [13].

2.3 | Echocardiographic Acquisition and Analysis

Transthoracic echocardiography was conducted using GE Vivid E90 systems (GE Healthcare, Chicago, IL, USA) with 4Vc and M5Sc transducers. All echocardiographic measurements strictly adhered to american society of echocardiography (ASE) guidelines [14].

Conventional parameters. Left and right ventricular dimensions, mitral doppler parameters, tissue doppler e′, and RV systolic velocity at the lateral tricuspid annulus S′ were measured from three consecutive cardiac cycles and averaged to minimize beat-to-beat variability. Pulmonary artery systolic pressure (PASP) was estimated from tricuspid regurgitation velocity using the modified Bernoulli equation and right atrial pressure derived from inferior vena cava diameter and collapsibility.

Two–dimensional RV strain. RV focused apical four-chamber views were exported in DICOM format and offline analysis using EchoPAC software (version 204, GE Healthcare). The endocardial borders of the RV free wall were manually traced at end-systole, and the software automatically generated global RV free–wall longitudinal strain (RVFWLS) values. Strain values are reported

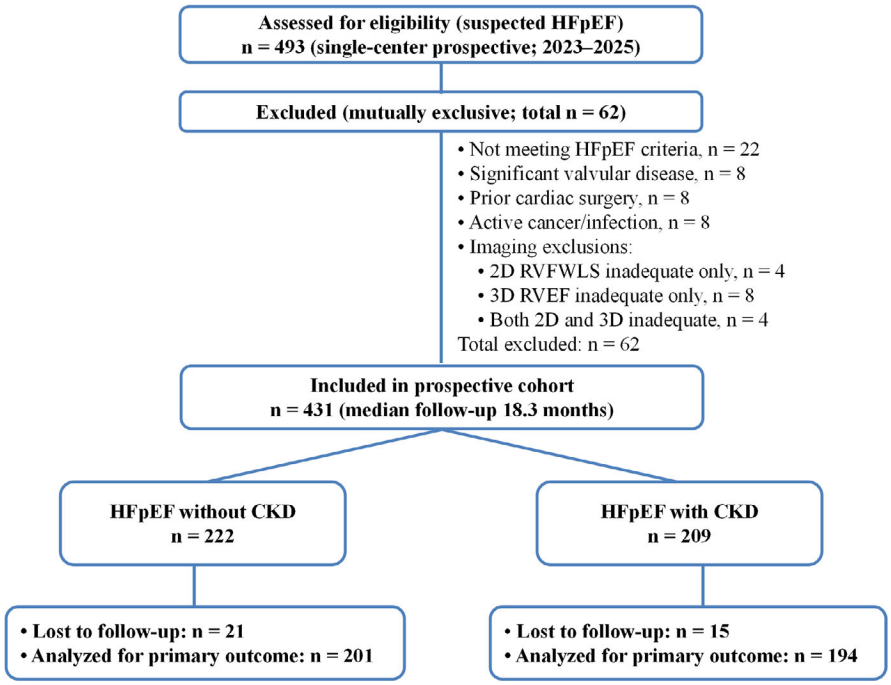


FIGURE 1 | **Patient flowchart**.
Total numbers and reasons for exclusion are detailed at each step. HFpEF, heart failure with preserved ejection fraction; CKD, chronic kidney disease.

as absolute magnitudes for clarity (higher values indicate better function).

Three-dimensional RV function. Full-volume 3D datasets were obtained from apical windows using a four-beat ECG–gated breath-hold protocol. The datasets were analyzed using EchoPAC 4D Auto RVQ software to quantify RV end-diastolic volume (RVEDV) and RV end-systolic volume (RVESV), which were indexed to body surface area (RVEDVi and RVESVi) according to the Mosteller formula. RV ejection fraction (RVEF) was calculated as (RVEDV − RVESV) / RVEDV × 100%. Representative imaging examples are shown in Figure 2.

To minimize the influence of vendor dependency—a known limitation in strain imaging—all image acquisition, 2D strain, and 3D volumetric analyses were uniformly performed using the single GE commercial platform and associated software versions specified above.

2.4 | Reproducibility

We reassessed 40 randomly selected subjects (approximately 10%) to examine measurement reproducibility. The same observer repeated all parameters at least 4 weeks later, blinded to the prior results and outcomes, and independently reselected the original echocardiographic images for 2D RV strain parameters and 3D RVEF. A second trained observer, unaware of the first observer's results and of all outcomes, independently repeated the measurements on the original echocardiographic images. Intraclass correlation coefficients (ICCs) and the Bland–Altman analyses were conducted for all parameters. An ICC value > 0.75 was considered indicative of good agreement, and coefficients of variation (CVs) <10% were deemed acceptable.

2.5 | Endpoints and Follow–up

The primary outcome was HF rehospitalization or CV death. HF rehospitalization was defined as an unplanned admission due to worsening HF symptoms or signs requiring intravenous diuretics, vasodilators, or other escalation of HF therapy. CV death included death attributable to HF, myocardial infarction, fatal arrhythmia, stroke, or other clearly documented cardiovascular causes.

Follow-up was conducted according to a standardized protocol at 3, 6, 12, and 18 months, then every 6 months by telephone or outpatient visit. Follow-up data were collected through review of electronic medical records, scheduled clinic evaluations, and structured telephone interviews. Patients without an event were censored at the date of their last confirmed follow-up. The data cutoff date was October 30, 2025.

All endpoint events were adjudicated by two senior cardiologists blinded to echocardiographic and clinical results. In case of disagreement, consensus was reached after joint review or, if necessary, by consultation with a third independent adjudicator. Event classification was finalized only after full verification of source documentation.

2.5.1 | Statistical Analysis

Missing data were assessed before analysis. Variables with <5% missingness were handled using complete-case analysis without imputation. Continuous data are presented as mean ± SD or median (interquartile range (*IQR*)) and compared using Student's *t*–test or the Mann–Whitney *U* test as appropriate. Categorical variables are summarized as *n* (%) and compared using χ^2 or the Fisher's exact tests.

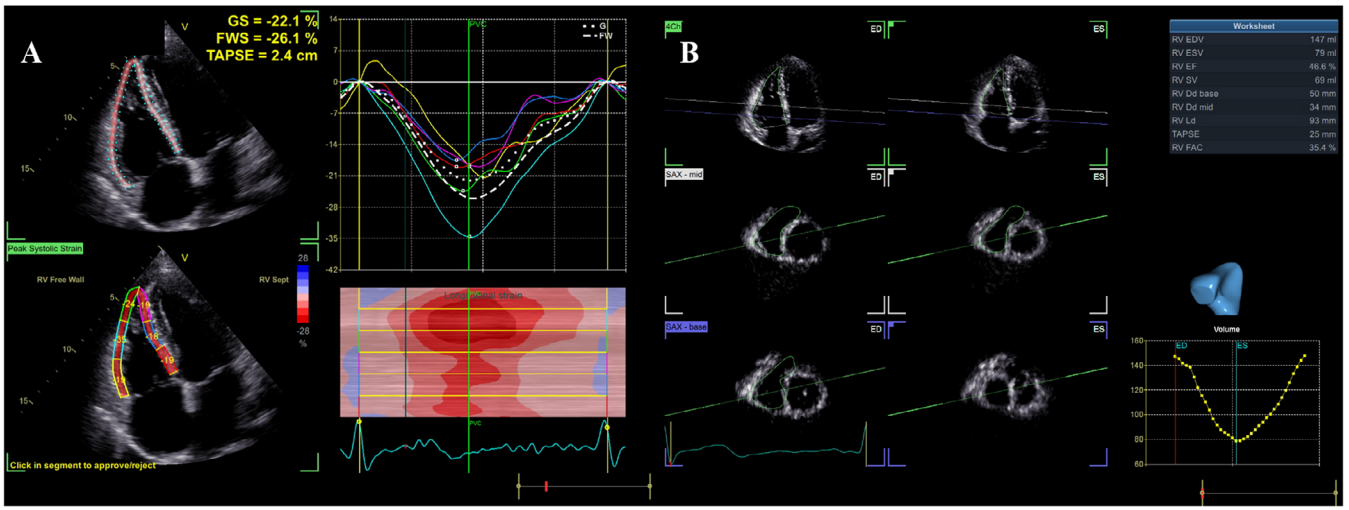


FIGURE 2 | **Two-dimensional speckle–tracking and three-dimensional echocardiographic evaluation of right ventricular function in HFpEF.**

A, 2D—RVFWLS; B, 3D echocardiographic of RVEDV、RVESV、RVEF.

Time-to-event outcomes were analyzed using the cox proportional hazards models, model 1: age, sex, BMI, eGFR, NT–proBNP, HDL–C, RV S′, mitral E/A, Model 2 (Model 1 + TyG), Model 3 (Model 1 + RVFWLS), Model 4 (Model 1 + 3D–RVEF), Model 5 (Model 1 + TyG + 3D–RVEF), and Model 6 (Model 1 + TyG + 2D–RVFWLS). All variance inflation factors (VIF) were < 3.

Unadjusted survival curves for the composite endpoint were displayed with the Kaplan–Meier curves and compared by the log–rank test, with numbers at risk shown. Where indicated, we plotted inverse-probability-weighted (IPW) adjusted survival curves and used weighted log–rank tests.

Model performance was evaluated using Harrell's concordance index (C–index) and 95% bias–corrected bootstrap confidence intervals (1000 resamples) [15]. Optimism-corrected C–indices were derived by internal bootstrap validation. Model fit was compared using the likelihood–ratio test and Akaike information criterion (AIC). Discrimination and reclassification were further examined using the category-free net reclassification improvement (cfNRI) and integrated discrimination improvement (IDI). Calibration was assessed by calibration plots, and clinical utility was evaluated using decision curve analysis (DCA).

For the HF rehospitalization component, we fitted Fine–Gray subdistribution hazards models, treating all–cause death prior to rehospitalization as the competing event [16]. The Cumulative incidence functions were compared with Gray's test. The proportional subdistribution hazards assumption was evaluated using covariate × log (time) interaction terms and graphical diagnostics. The Fine–Gray models were adjusted for the same covariates as the cox models (Model 1–Model 6). The composite endpoint (CV death or HF rehospitalization) was analyzed only with cox models; proportional hazards were assessed with Schoenfeld residuals and global tests.

To ensure model stability, the ratio of events per variable (EPV) was maintained at ≥ 10 for all multivariable models. Sensitivity analyses were conducted across prespecified subgroups (sex, BMI category, and eGFR strata) to confirm robustness and generalizability of the main findings.

All statistical analyses were performed using SPSS (version 26.0, IBM Corp., Armonk, NY, USA), R software (version 4.4.2), and Python (version 3.10). Two–tailed $p < 0.05$ was considered statistically significant.

3 | Results

3.1 | Baseline Characteristics and Echocardiographic Characteristics

A total of 395 patients with HFpEF were enrolled, of whom 194 (49.1%) had concomitant CKD. The median age was 65 years (*IQR* 54–73), and 267(67.5%) were women. Compared with non-CKD patients, those with CKD had a lower eGFR (43.3 [32.2–54.3] vs. 87.1 [80.3–101.4] mL/min/1.73m^2, $P < 0.001$), higher serum creatinine (131.9 [103.7–184.0] vs. 88.5 [71.3–103.1] μmol/L, $P < 0.001$), elevated TyG index (9.4 [8.9–9.9] vs. 8.7 [8.3–9.3], $P < 0.001$), higher NT–proBNP (1370.1 [803.1–2,217.5] vs. 514.1 [376.1–714.0], $P < 0.001$), and higher FBG (6.3 [4.9–8.3] vs 5.5 [4.8–6.6] mmol/L, $P = 0.001$). Other demographics, comorbidities, and medications were similar. Characteristics are summarized in Table 1, and stratified findings by TyG tertiles in Table S1.

Conventional echocardiographic parameters were comparable between groups, including left ventricular dimensions, diastolic parameters, and RV parameters (Table 2). However, advanced RV imaging demonstrated more pronounced RV dysfunction in CKD. Compared with non–CKD, patients with CKD had lower absolute RVFWLS (16.7 (12.4, 19.2) versus 19.5 (17.6, 21.6) %, $p < 0.001$; higher absolute values indicate better function), higher RVESVi (33.0 [26.4–41.7] versus 29.3 [24.0–36.9] mL/m^2, $P < 0.001$), and lower 3D–RVEF (45.3 (40.0–50.1) versus 48.8 (44.5–52.9) %, $P < 0.001$; Table 3)

TABLE 1 | Baseline characteristics of patients with HFpEF stratified by CKD status.

Variables	All patients (N = 395)	HFpEF without CKD (n = 201)	HFpEF with CKD (n = 194)	*p* value	Std diff
Age, years	65.0 (54.0, 73.0)	64.0 (54.0, 73.0)	65.5 (55.0, 71.8)	0.942	0.010
Sex, female, n (%)	267 (67.6)	135 (67.2)	132 (68.0)	0.937	0.027
BMI, kg/m^2	23.2 (20.5, 25.9)	23.5 (20.6, 26.6)	22.9 (20.5, 25.1)	0.136	0.183
NYHA class, n (%)				**<0.001**	
II	221 (56.0)	136 (67.7)	85 (43.8)		0.495
III	138 (34.9)	57 (28.4)	81 (41.8)		0.284
IV	36 (9.1)	8 (4.0)	28 (14.4)		0.368
SBP, mmHg	135.0 (118.5, 150.5)	134.0 (121.0, 150.0)	136.0 (114.0, 152.8)	0.806	0.036
DBP, mmHg	82.0 (73.0, 91.0)	82.0 (74.0, 90.0)	82.0 (73.0, 91.0)	0.746	0.019
Current or former smoker, n (%)	113 (28.6)	54 (26.9)	59 (30.4)	0.436	0.078
Comorbidities, n(%)					
Hypertension	289 (73.2)	145 (72.14)	144 (74.2)	0.640	0.047
Dyslipidemia	54 (13.7)	26 (12.9)	28 (14.4)	0.665	0.044
Diabetes	176 (44.6)	82 (40.8)	94 (48.4)	0.126	0.155
Coronary artery disease	121 (30.6)	64 (31.8)	57 (29.4)	0.596	0.053
Atrial fibrillation/flutter	160 (40.5)	76 (37.8)	84 (43.3)	0.267	0.272
Medication use, n(%)					
SGLT-2i	217 (54.9)	105 (52.2)	112 (57.7)	0.273	0.111
Insulin	93 (23.5)	43 (21.4)	50 (25.8)	0.364	0.103
β–Blockers	242 (61.3)	115 (57.2)	127 (65.5)	0.092	0.170
ACEI/ARB	226 (57.2)	113 (56.2)	113 (58.2)	0.684	0.041
CCB	148 (37.5)	73 (36.3)	75 (38.6)	0.631	0.163
Aspirin	195 (49.4)	107 (53.2)	88 (45.4)	0.118	0.158
Diuretics	295 (74.7)	144 (71.6)	151 (77.8)	0.157	0.143
Statin	316 (80.0)	158 (78.6)	158 (81.4)	0.481	0.071
Laboratory test					
Serum creatinine, μmol/L	103.1 (81.0, 141.6)	88.5 (71.3, 103.1)	131.9 (103.7, 184.0)	**<0.001**	1.166
eGFR, mL/min/1.73m^2	71.2 (40.5, 89.6)	87.1 (80.3, 101.4)	43.3 (32.2, 54.3)	**<0.001**	1.712
NT–proBNP	741.9 (451.7, 1457.7)	514.1 (376.1, 714.0)	1370.1 (803.1, 2217.5)	**<0.001**	1.337
Total cholesterol, mmol/L	3.9 (3.1, 4.7)	3.9 (3.1, 4.7)	3.8 (3.1, 4.7)	0.611	0.071
Triglycerides, mmol/L	1.5 (1.0, 2.0)	1.4 (1.0, 1.8)	1.5 (1.2, 2.1)	0.058	0.011
Fasting blood glucose, mmol/L	5.8 (4.8, 7.3)	5.5 (4.8, 6.6)	6.3 (4.9, 8.3)	**0.001**	0.326
HDL–C, mmol/L	0.9 (0.7, 1.0)	0.9 (0.7, 1.1)	0.8 (0.7, 1.0)	0.067	0.218
LDL–C, mmol/L	2.1 (1.6, 2.8)	2.1 (1.6, 2.8)	2.1 (1.4, 2.7)	0.157	0.171
TyG index	9.0 (8.5, 9.6)	8.7 (8.3, 9.3)	9.4 (8.9, 9.9)	**<0.001**	0.714
Follow–up outcomes, n(%)					
Composite endpoint	158 (40.0)	45 (22.4)	113 (58.3)	**<0.001**	0.785
CV death	39 (9.9)	8 (4.0)	31 (16.0)	**<0.001**	0.409
HF rehospitalization	119 (30.1)	37 (18.4)	82 (42.3)	**<0.001**	0.537

Continuous variables are expressed as median (interquartile range), and categorical variables as number (percentage). *P* values were obtained using the Mann–Whitney U test for continuous variables and the χ^2 test (or Fisher's exact test when appropriate) for categorical variables. Standardized differences (Std diff) were

(Continues)

TABLE 1 | (Continued)

calculated for all variables to assess baseline comparability between groups. Values ≥0.10 indicate potentially meaningful imbalance between groups. Chronic kidney disease (CKD) was defined as an eGFR <60 mL/min/1.73 m^2. Abbreviations: ACEI/ARB = angiotensin–converting enzyme inhibitor/angiotensin II receptor blocker; BMI = body mass index; CCB = calcium channel blocker; CV = Cardiovascular; DBP = diastolic blood pressure; eGFR = estimated glomerular filtration rate; HDL–C = high density lipoprotein–cholesterol; LDL–C = low density lipoprotein–cholesterol; NT–proBNP = N–terminal pro–B–type natriuretic peptide; NYHA = New York Heart Association; SBP = systolic blood pressure; Scr = serum creatinine; TC = total cholesterol; TG = triglycerides; TyG = triglyceride glucose.

3.2 | Prognostic Models and Model Performance

During a median follow-up of 18.3 months, 158 (40.0%) patients reached the primary composite endpoint. The CKD group had higher rates of CV death (16.0% vs. 4.0%, $P < 0.001$) and HF rehospitalization (42.3% vs. 18.4%, $P < 0.001$). Multivariable cox models (Table 4) showed both TyG index (95% *CI* 1.26–1.92; $P < 0.001$) and 2D–RVFWLS (95% *CI* 0.88–0.95; $P < 0.001$) as independent predictors. 3D–RVEF showed a weaker association (95% *CI* 0.93–0.99; $P = 0.020$), but lost significance after full adjustment.

3.3 | Survival Analyses (unadjusted and IPW–adjusted)

The Kaplan–Meier survival curves worse event-free survival in CKD (log–rank $P<0.001$; Figure 3A), higher TyG tertiles (Figure 3B), and lower absolute 2D–RVFWLS values (Figure 3C), with a nonsignificant trend for lower 3D–RVEF (log–rank $p = 0.062$; Figure 3D). Combined TyG index and RVFWLS stratification provided greatest separation (overall log–rank $p < 0.001$; Figure 3E).

Inverse probability-weighted (IPW)—adjusted curves confirmed these associations (all $P<0.001$ except RVEF $p = 0.587$; Figures 3F–J).

The model combining TyG and 2D–RVFWLS achieved the best overall discrimination (optimism-corrected C–index 0.579 (apparent 0.735)) and improved model fit (ΔAIC: –31.69), with significant reclassification gains (cfNRI = 0.859; IDI = 0.749, both $p < 0.001$; Table 5). Calibration and decision curve analyses confirmed superior calibration and net benefit for this model (Figures 4 and 5).

3.4 | Competing–Risk and Subgroup Analyses

In Fine–Gray models for HF rehospitalization with all–cause death as the competing event, both the TyG index and absolute RVFWLS remained independent in the Model 6 (Table 6, Figure S1–S2).

3.5 | Reproducibility of RV Function

Reproducibility was excellent for both RVFWLS and RVEF. Intraobserver ICCs were 0.90 (RVFWLS) for both parameters (CVs 3.6 % and 2.9 %), and interobserver ICCs were 0.89 and 0.80 (CVs < 4 %). (Table S2 and Figure S3).

4 | Discussion

This study aimed to evaluate whether integrating RV function—measured by 2D–RVFWLS and 3D–RVEF—with the TyG index improves prognostic evaluation in HFpEF, with a particular focus on patients with CKD.

Our principal findings were as follows: (1) RV function, as measured by both 2D–RVFWLS and 3D–RVEF, was notably impaired in HFpEF patients with CKD, with 2D–RVFWLS offering a more reliable and practical prognostic indicator than 3D–RVEF; (2) the TyG index emerged as an independent predictor of adverse outcomes in HFpEF, where elevated levels were linked to a markedly higher risk of CV death or HF rehospitalization; (3) Combining TyG with 2D–RVFWLS significantly improved risk stratification in HFpEF patients, particularly in the CKD subgroup.

4.1 | HFpEF With CKD and RV Dysfunction

The mechanisms of exacerbated RV dysfunction in HFpEF with CKD are multifaceted, embodying bidirectional cardiorenal interactions. CKD increases RV afterload through systemic inflammation and endothelial dysfunction, promoting vascular remodeling and RV–pulmonary artery uncoupling. Sympathetic overactivation and volume overload increase right atrial pressure and venous congestion, amplifying RV preload and dilation. Metabolic insults such as hyperuricemia and anemia compromise RV oxygenation and energetics. These converge with neurohormonal surges, independent of left ventricular baseline. Our study findings align with these mechanistic hypotheses: In HFpEF patients with CKD, we observed substantial RV impairment, as quantified through 2D–RVFWLS and 3D–RVEF, with more pronounced damage in the CKD groups, reflecting the cumulative effects of the aforementioned pathways involving inflammation, and neurohormonal activation [17].

4.2 | Prognostic Role of the Triglyceride–Glucose (TyG) Index

The triglyceride–glucose (TyG) index, serves as a simple, validated surrogate marker for insulin resistance (IR), offering robust prognostic value in cardiovascular disorders, including HFpEF[20]. In HFpEF, where metabolic comorbidities like obesity and CKD are prevalent, IR—reflected by elevated TyG—plays a pivotal role in exacerbating RV dysfunction through interconnected pathophysiological pathways. Mechanistically, IR disrupts myocardial energy metabolism by impairing glucose uptake and oxidation while favoring fatty acid utilization, leading to lipotoxicity, mitochondrial dysfunction, and reduced ATP production in the RV, which is particularly susceptible due to its thinner wall and reliance on efficient energetics for handling variable pulmonary loads. This metabolic remodeling is compounded by IR–driven systemic inflammation via activation of pro-inflammatory cytokines, endothelial dysfunction, and

TABLE 2 | Conventional echocardiographic parameters in patients with HFpEF stratified by CKD.

Variables	All patients (N = 395)	HFpEF without CKD (n = 201)	HFpEF with CKD (n = 194)	*p* value	Std diff
Left ventricular structure					
LVEDd, mm	47.9 (44.7, 51.4)	47.6 (44.7, 51.4)	48.0 (44.7, 51.3)	0.705	0.012
IVSd, mm	11.0 (10.0, 11.9)	11.0 (9.9, 11.8)	11.0 (10.0, 12.0)	0.269	0.091
LVPWd, mm	10.0 (9.0, 11.0)	10.0 (9.0, 11.0)	10.0 (9.0, 11.0)	0.973	0.020
LVMI, g/m^2	103.6 (89.0, 124.5)	102.9 (87.5, 123.1)	105.3 (89.3, 125.8)	0.358	0.062
Left ventricular diastolic and systolic function					
LAVI, mL/m^2	36.1 (34.9, 38.1)	36.2 (34.9, 38.0)	36.1 (34.8, 38.1)	0.273	0.995
Mitral E, cm/s	68.3 (58.9, 83.0)	65.4 (55.8, 73.8)	62.0 (55.7, 70.8)	0.085	0.291
Mitral A, cm/s	81.0 (67.9, 89.6)	79.0 (65.0, 91.8)	82.0 (70.8, 88.7)	0.824	0.038
Septal e', cm/s	5.5 (4.5, 6.3)	5.5 (4.4, 6.4)	5.5 (4.5, 6.3)	0.867	0.035
Mitral E/A	0.7 (0.6, 0.9)	0.7 (0.6, 1.0)	0.7 (0.6, 0.9)	0.116	0.158
E/e'	11.8 (10.0, 13.8)	11.8 (10.0, 14.1)	11.8 (9.9, 13.6)	0.484	0.117
LVEF,%	56.3 (54.0, 58.7)	56.3 (54.0, 59.0)	56.3 (53.7, 58.5)	0.284	0.149
Right ventricular parameters					
Tricuspid E, cm/s	55.1 (47.8, 62.0)	55.0 (47.4, 62.1)	55.2 (48.9, 61.8)	0.600	0.068
Tricuspid A, cm/s	31.4 (24.4, 38.8)	30.8 (24.8, 38.4)	32.7 (23.6, 39.1)	0.815	0.017
Tricuspid E/A	2.8 (2.4, 3.1)	2.8 (2.4, 3.1)	2.8 (2.5, 3.1)	0.600	0.069
TAPSE, mm	20.0 (18.0, 22.0)	20.0 (18.0, 23.0)	20.0 (18.0, 21.0)	0.062	0.000
RV S', cm/s	10.4 (9.1, 12.4)	10.6 (9.2, 12.8)	10.3 (9.0, 12.0)	0.099	0.142

Continuous variables are expressed as median (interquartile range). *P* values were derived using the Mann–Whitney U test.std diff ≥ 0.10 denotes meaningful imbalance.

Abbreviations:LVEDd = left ventricular end–diastolic diameter; E/e′ = ratio of mitral E velocity to early diastolic annular e′ velocity; IVSd = interventricular septal wall thickness at end–diastole; LAVI = left atrial volume index; LVMI = left ventricular mass index; LVPWd = left ventricular posterior wall thickness at end–diastole; RV S′ = systolic velocity at the lateral tricuspid annulus; TAPSE = tricuspid annular plane systolic excursion.

TABLE 3 | Right ventricular function derived from 2D strain and 3D echocardiography stratified by CKD status.

Variables	All patients (N = 395)	HFpEF without CKD (n = 201)	HFpEF with CKD (n = 194)	*p* value	Std diff
2D strain parameters					
2D–RVFWLS (absolute value, %)	18.3 (14.2, 20.9)	19.5 (17.6, 21.6)	16.7 (12.4, 19.2)	**<0.001**	0.675
3D structural and functional parameters					
RVEDV, mL	101.0 (83.0, 130.6)	100.0 (81.0, 123.0)	102.6 (84.2, 135.5)	0.181	0.146
RVEDVi, mL/m^2	58.1 (47.6, 74.9)	56.3 (47.5, 70.8)	60.3 (47.7, 78.6)	0.073	0.191
RVESV, mL	54.4 (44.0, 67.8)	51.1 (41.0, 65.4)	58.4 (46.5, 70.5)	**<0.001**	0.334
RVESVi, mL/m^2	31.2 (24.9, 38.9)	28.8 (23.8, 36.1)	33.4 (27.0, 41.8)	**<0.001**	0.371
RVEF, %	46.9 (42.7, 51.3)	48.8 (44.5, 52.9)	45.3 (40.0, 50.1)	**<0.001**	0.702

Continuous variables are presented as median (interquartile range). *P* values were derived using the Mann–Whitney U test. Std diff ≥ 0.10 denotes meaningful imbalance. For absolute RVFWLS, higher absolute RVFWLS indicates better systolic function.

Abbreviations: RVEDV = right ventricular end–diastolic volume; RVEDVi = right–ventricular end–diastolic volume indexed to body–surface area; RVEF = right ventricular ejection fraction measured by three-dimensional echocardiography; RVESV = right ventricular end–systolic volume; RVESVi = right ventricular end–systolic volume indexed to body–surface area; RVFWLS = right ventricular free wall longitudinal strain.

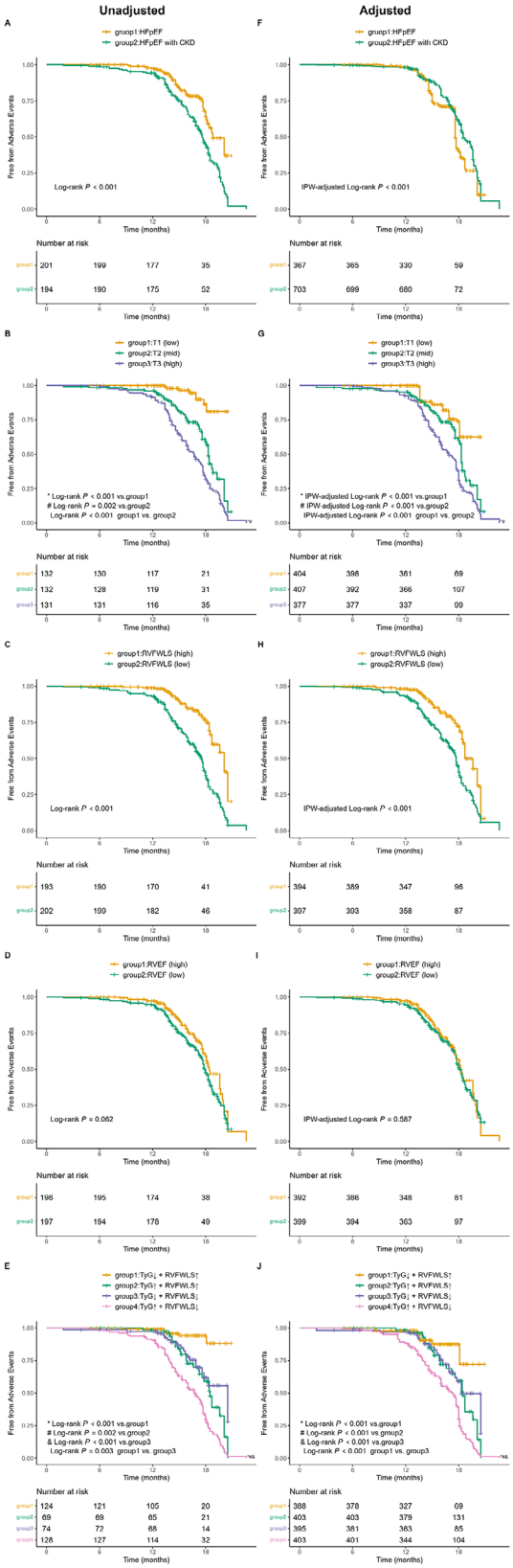


FIGURE 3 | Survival curves showing the association of CKD status, TyG index, and RV parameters with outcomes.

Kaplan–Meier survival curves for the composite endpoint (cardiovascular death or HF rehospitalization) in HFpEF patients.

Panels A–E: Unadjusted curves stratified by CKD status (A), TyG index tertiles (B), RVFWLS dichotomized by median (C), RVEF dichotomized by median (D), and combined TyG index (high/low) with RVFWLS (high/low) (E). Panels F–J: Corresponding inverse probability weighting (IPW)–adjusted curves, accounting for age, sex, BMI, eGFR, log(NT–proBNP), HDL–C, RV S′, and mitral E/A. Log–rank *p* values are shown. Numbers at risk are displayed below each curve. Abbreviations: CKD = chronic kidney disease; TyG = triglyceride–glucose index; RV = right ventricle; RVFWLS = right ventricular free–wall longitudinal strain; RVEF = right ventricular ejection fraction; HFpEF = heart failure with preserved ejection fraction; BMI = body mass index; eGFR = estimated glomerular filtration rate; NT–proBNP = N–terminal pro–B–type natriuretic peptide; HDL–C = high–density lipoprotein cholesterol.

oxidative stress, which promote pulmonary vascular remodeling [18] and heightened RV afterload, culminating in RV–pulmonary artery uncoupling and contractile impairment. Furthermore, IR accelerates extracellular matrix deposition and myocardial fibrosis through pathways like TGF–β signaling and advanced glycation end products, stiffening the RV myocardium and impairing diastolic relaxation—hallmarks of HFpEF progression. In the context of concomitant CKD, these effects are amplified by uremic toxins, renal-mediated volume overload, and enhanced cardiorenal–neurohormonal crosstalk, further sensitizing the RV to IR-induced vulnerability. Our study corroborates these links, revealing that elevated TyG independently predicted cardiovascular mortality or heart failure rehospitalization in HFpEF patients, with enhanced discriminative power in the CKD subgroup [21].

4.3 | Prognostic Value of RV Function and TyG Index in HFpEF With CKD

Previous studies have consistently shown that patients with HFpEF and CKD exhibit more severe disease phenotypes and poorer clinical outcomes, highlighting the adverse impact of CKD and the necessity for optimized renal management [22]. Integrating RV functional indicators with the TyG index can provide complementary information, as RV function reflects hemodynamics and pulmonary vascular remodeling, while TyG captures metabolic stress associated with insulin resistance; their combination can more comprehensively capture the mechanisms by which CKD exacerbates cardiorenal–pulmonary axis injury. In this study, the model combining 2D–RVFWLS and TyG demonstrated superior performance, including significant improvements in the optimism-corrected C–index, reclassification enhancements in cfNRI (0.859) and IDI (0.749) (both $P < 0.001$), and good model calibration. Compared with single indicators or models combining 3D–RVEF, this integrated model showed higher ΔC–index (e.g., ΔC–index = +0.061 compared to the base model) and net benefit under decision curve analysis, particularly at moderate to high-risk thresholds, underscoring the superiority of 2D–RVFWLS in practicality and predictive accuracy. Robustness validation further supports this finding. The superiority of 2D-RVFWLS over 3D-RVEF as a prognostic predictor can be attributed to the fundamental difference in their measurement: Strain is a more sensitive marker of subclinical myocardial deformation and intrinsic contractility, capable of detecting myocardial stiffening and fibrosis that precedes a measurable decline in global function (EF) [19]. Furthermore, volumetric indices like 3D–RVEF are highly susceptible to complex loading conditions, while strain is relatively less load-

TABLE 4 | Multivariable cox regression models for the composite endpoint: Incremental value of TyG index and RV parameters.

Variables	Model 1 (Base)		Model 2 (+TyG)		Model 3 (+RVEF)		Model 4 (+RVFWLS)		Model 5 (+TyG+RVEF)		Model 6 (+TyG+RVFWLS)	
	HR (95%CI)	*P*	HR (95%CI)	*P*	HR (95%CI)	*P*	HR (95%CI)	*P*	HR (95%CI)	*P*	HR (95%CI)	*P*
Age, years	0.99 (0.98–1.00)	0.131	0.99 (0.98–1.00)	**0.029**	0.99 (0.98–1.00)	0.175	0.99 (0.98–1.00)	0.127	0.99 (0.98–1.00)	0.046	0.99 (0.98–1.00)	0.043
Sex, female	0.78 (0.56–1.10)	0.154	0.80 (0.57–1.12)	0.192	0.79 (0.56–1.10)	0.164	0.77 (0.55–1.09)	0.142	0.80 (0.57–1.12)	0.186	0.77 (0.54–1.08)	0.130
BMI, kg/m^2	0.97 (0.93–1.02)	0.223	0.97 (0.93–1.01)	0.181	0.97 (0.93–1.02)	0.218	0.97 (0.93–1.02)	0.282	0.97 (0.93–1.02)	0.212	0.97 (0.93–1.02)	0.273
eGFR, ml/min/1.73m^2	0.99 (0.98–1.00)	**0.001**	0.99 (0.98–1.00)	**0.037**	0.99 (0.98–1.00)	**0.003**	0.99 (0.98–1.00)	**0.023**	0.99 (0.98–1.00)	0.053	0.99 (0.99–1.00)	0.155
log(NT–proBNP)	1.32 (1.01–1.73)	**0.038**	1.20 (0.92–1.58)	0.185	1.30 (0.99–1.70)	0.058	1.26 (0.97–1.66)	0.087	1.17 (0.89–1.54)	0.263	1.20 (0.91–1.57)	0.199
HDL–C, mmol/L	0.40 (0.20–0.79)	**0.008**	0.55 (0.28–1.11)	0.094	0.39 (0.20–0.79)	**0.008**	0.41 (0.21–0.81)	**0.010**	0.54 (0.27–1.07)	0.078	0.50 (0.25–1.00)	0.048
RV S', cm/s	1.00 (0.92–1.07)	0.905	1.01 (0.93–1.09)	0.892	1.01 (0.93–1.09)	0.895	1.00 (0.92–1.08)	0.978	1.02 (0.94–1.10)	0.637	1.00 (0.93–1.09)	0.923
Mitral E/A	0.53 (0.31–0.89)	**0.016**	0.50 (0.29–0.84)	**0.009**	0.56 (0.33–0.94)	**0.028**	0.55 (0.32–0.93)	**0.026**	0.54 (0.32–0.91)	0.019	0.53 (0.31–0.90)	**0.019**
TyG index	—	—	1.56 (1.26–1.92)	**<0.001**	—	—	—	—	1.57 (1.27–1.94)	**<0.001**	1.41 (1.14–1.75)	**0.001**

(Continues)

TABLE 4 | (Continued)

Variables	Model 1 (Base)		Model 2 (+TyG)		Model 3 (+RVEF)		Model 4 (+RVFWLS)		Model 5 (+TyG+RVEF)		Model 6 (+TyG+RVFWLS)	
	HR (95%CI)	*P*	HR (95%CI)	*P*	HR (95%CI)	*P*	HR (95%CI)	*P*	HR (95%CI)	*P*	HR (95%CI)	*P*
RVEF, %	—	—	—	—	0.96 (0.93–1.00)	**0.028**	—	—	0.96 (0.93–0.99)	0.020	—	—
RVFWLS, %	—	—	—	—	—	—	0.90 (0.87–0.94)	**<0.001**	—	—	0.91(0.88–0.95)	**<0.001**
Model performance												
AIC	1501.88		1486.56		1499.15		1479.08		1483.79		1470.19	
Apparent C-index	0.674		0.711		0.677		0.713		0.714		0.735	
.632+ C–index (95 % CI)	0.556 (0.501–0.612)		0.569 (0.514–0.625)		0.559 (0.506–0.614)		0.573 (0.525–0.627)		0.570 (0.522–0.623)		0.579 (0.522–0.633)	
ΔC–index vs Model 1			+0.037		+0.003		+0.039		+0.040		+0.061	
ΔAIC vs Model 1			−15.32		−2.73		−22.80		−18.09		−31.69	
LR χ^2(df), *P*			17.32(1), **<0.001**		4.73(1), **0.035**		24.79(1). **<0.001**		22.09(2), **<0.001**		35.68(2), **<0.001**	

HRs were derived from Cox proportional hazards models. Model 1 (Base) included age, sex, BMI, eGFR, NT-proBNP, HDL-C, RV s′, and Mitral E/A. Model 2 = Base + TyG index; Model 3 = Base + RVEF; Model 4 = Base + RVFWLS; Model 5 = Base + TyG index + RVEF; Model 6 = Base + TyG index + RVFWLS (final model). RVFWLS was analyzed as an absolute value, higher absolute RVFWLS indicates better systolic function. Apparent C-index was optimism-corrected using the .632+ bootstrap (1 000 resamples). LR χ^2 compares each model with Model 1. AIC denotes Akaike information criterion.—, variable not included in model.

TABLE 5 | Paired .632+ bootstrap comparison of model reclassification at 18 months (cfNRI, IDI).

Comparison	cfNRI (95 % CI)	*P* (cfNRI)	IDI (95 % CI)	*P* (IDI)
Model 2 vs Model 1	0.657 (0.465–0.852)	**<0.001**	0.310 (0.181–0.429)	**<0.001**
Model 3 vs Model 1	0.186 (−0.012–0.385)	**0.035**	0.097 (0.045–0.151)	**<0.001**
Model 4 vs Model 1	0.592 (0.392–0.777)	**<0.001**	0.536 (0.367–0.713)	**<0.001**
Model 5 vs Model 1	0.665 (0.461–0.857)	**<0.001**	0.437 (0.302–0.580)	**<0.001**
Model 6 vs Model 1	0.859 (0.674–1.048)	**<0.001**	0.749 (0.565–0.939)	**<0.001**
Model 4 vs Model 3	0.427 (0.226–0.639)	**<0.001**	0.440 (0.259–0.629)	**<0.001**
Model 5 vs Model 2	0.284 (0.105–0.478)	**0.001**	0.127 (0.065–0.189)	**<0.001**
Model 6 vs Model 5	0.289 (0.092–0.495)	**0.004**	0.312 (0.147–0.487)	**<0.001**

cfNRI, and IDI were derived from 1 000 paired .632+ bootstrap samples for 18-month composite endpoint. cfNRI quantifies the proportion of subjects whose predicted risk moved in the desirable direction without the need for predefined risk cut-points; IDI reflects the change in the mean predicted probabilities between events and non-events. Model 1 (Base) included age, sex, BMI, eGFR, NT-proBNP, HDL-C, RV s′, and Mitral E/A. Model 2 = Base + TyG index; Model 3 = Base + RVEF; Model 4 = Base + RVFWLS; Model 5 = Base + TyG index + RVEF; Model 6 = Base + TyG index + RVFWLS (final model). Positive ΔC-index, cfNRI, or IDI values together with $P < 0.05$ indicate that the new model discriminates events from non-events and/or reclassifies patients more accurately than the comparison model.

Abbreviations: cfNRI: category–free Net Reclassification Improvement; CI: Confidence Interval; IDI: Integrated Discrimination Improvement.

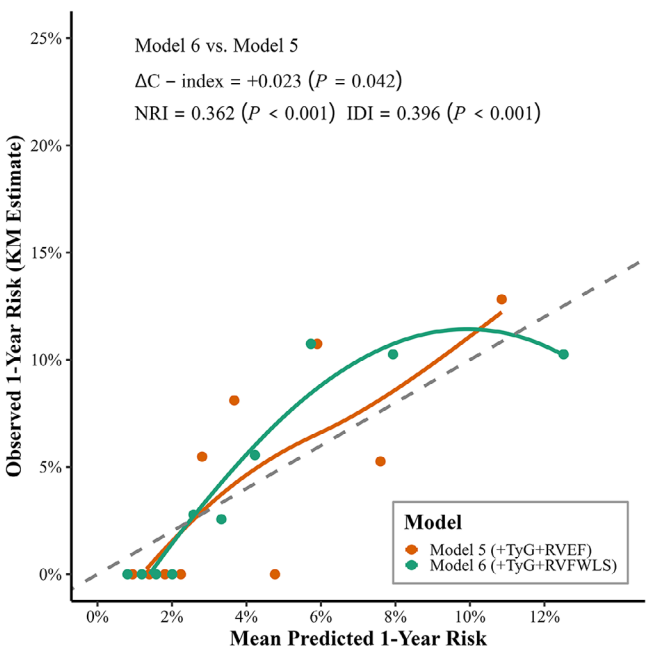


FIGURE 4 | Calibration plot at 1 year for prognostic models 5 and 6. Calibration curves for predicting 1–year risk of the composite endpoint in HFpEF patients. Each point represents a decile group of predicted risk. The dashed diagonal line indicates perfect calibration. Model 5 (TyG index + RVEF; green line) and Model 6 (TyG index + RVFWLS; red line) are shown, with Model 6 demonstrating closer agreement to the ideal line. See Table 4 for model details. Abbreviations: HFpEF = heart failure with preserved ejection fraction; TyG = triglyceride–glucose index; RVEF = right ventricular ejection fraction; RVFWLS = right ventricular free–wall longitudinal strain.

dependent, thus offering a more specific assessment of intrinsic RV function. The value of this integration lies in achieving better risk stratification, particularly for high-risk identification in CKD subgroups, and future multicenter studies should confirm its applicability in larger cohorts.

4.4 | Therapeutic Implications: Residual Metabolic Risk beyond GDMT

The prognostic independence of the TyG index carries significant therapeutic implications, especially in the era of guideline-directed medical therapy (GDMT) for HFpEF. SGLT2 inhibitors (SGLT2i), which exert benefits partly through metabolic mechanisms, are now foundational in HFpEF treatment. However, our finding that the TyG index remained a significant, independent predictor even after adjusting for SGLT2i usage suggests it captures residual cardiometabolic risk not fully mitigated by current standard therapy. This points toward the need for more targeted metabolic intervention in high-risk patients. For example, agents like GLP-1 receptor agonists, which have demonstrated compelling cardiovascular and metabolic benefits—particularly in patients with obesity and Type 2 Diabetes—may offer a synergistic approach. The combination of RV strain analysis (for functional risk) and the TyG index (for metabolic risk) could help identify a precise phenotype of patients who would benefit most from adding an aggressive metabolic modulator like a GLP-1 receptor agonist to their existing SGLT2i regimen.

4.5 | Clinical Implications

Our findings highlight the potential of integrating metabolic and functional markers into routine risk stratification for HFpEF, particularly when CKD is present. Because both the TyG index and 2D–RVFWLS are inexpensive, noninvasive, and widely accessible, their combined use could provide a pragmatic bedside tool to identify high-risk patients who might otherwise appear stable based on conventional echocardiographic or biochemical indices alone. In particular, HFpEF patients with concomitant CKD and combined metabolic–RV impairment represent a vulnerable phenotype that may warrant intensified surveillance and targeted therapeutic interventions. Furthermore, serial assessment of TyG and RV strain may serve as responsive surrogate endpoints in monitoring treatment efficacy, offering opportunities for precision medicine approaches in this complex population.

TABLE 6 | Fine–Gray competing risk models for heart failure rehospitalization: Incremental value of TyG index and RV function.

Variables	Model 1 (Base)		Model 2 (+TyG)		Model 3 (+RVEF)		Model 4 (+RVFWLS)		Model 5 (+TyG+RVEF)		Model 6 (+TyG+RVFWLS)	
	SHR (95%CI)	*P*	SHR (95%CI)	*P*	SHR (95%CI)	*P*	SHR (95%CI)	*P*	SHR (95%CI)	*P*	SHR (95%CI)	*P*
TyG index		–	1.56(1.23–1.97)	**<0.001**					1.56(1.23–1.96)	**<0.001**	1.48(1.16–1.88)	**0.001**
RVEF,%		–	–	–	0.96(0.93–1.00)	0.054	–	–	0.96(0.93–1.00)	0.050		
RVFWLS,%		–	–	–	–	–	0.93(0.89–0.97)	**0.001**	–	–	0.94(0.90–0.98)	**0.003**
C–index	0.623(0.545–0.701)		0.659(0.584–0.735)		0.623(0.545–0.701)		0.666(0.591–0.741)		0.65(0.574–0.726)		0.685(0.612–0.758)	

SHRs were estimated using Fine–Gray competing risk models with HF rehospitalization as the event of interest and all–cause death as the competing event. Continuous variables were modeled as follows: TyG index (per 1-unit increase), RVFWLS (per 1% absolute value increase), and RVEF (per 1% increase). All models were adjusted for the same covariates as in Table 4 (age, sex, BMI, eGFR, NT-proBNP, HDL-C, RV s′, and Mitral E/A). Apparent C-indices (time-dependent at 18 months) were optimism-corrected using the .632+ bootstrap method (1,000 resamples). ΔC-index indicates the absolute increase versus Model 1. cfNRI and IDI are category–free and were calculated at 18 months with paired bootstrap resampling. Positive ΔC-index, cfNRI or IDI with $p < 0.05$ indicate improved model performance vs Model 1.SHR = subdistribution hazard ratio; CI = confidence interval; AIC = Akaike information criterion.

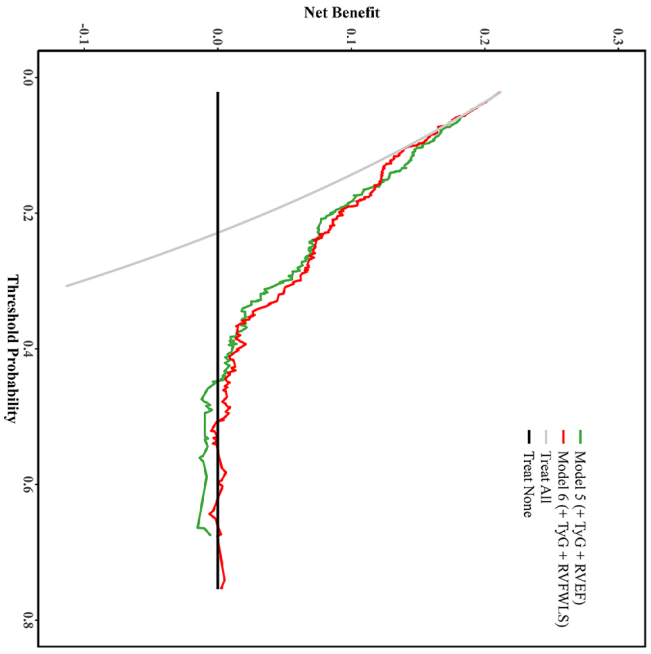


FIGURE 5 | Decision curve analysis of prognostic models in HFpEF. Decision curve analysis evaluating the net benefit of prognostic models across threshold probabilities (0.05–0.30) for the composite endpoint. The solid colored lines represent Models 1–6, with the green line denoting Model 5 (TyG index + 3D–RVEF) and the red line denoting Model 6 (TyG index + 2D–RVFWLS). The gray line indicates "treat all," and the black line indicates "treat none." Model 6 shows superior net benefit in clinically relevant ranges. See Table 4 for model details. Abbreviations: HFpEF = heart failure with preserved ejection fraction; TyG = triglyceride–glucose index; RVEF = right ventricular ejection fraction; RVFWLS = right ventricular free–wall longitudinal strain.

5 | Study Limitations

This study has several limitations. First, it is a single-center, prospective analysis, which may introduce selection bias and limit generalizability. We will conduct further multicenter validation to confirm the findings of this study. Second, although we adjusted for recognized covariates, residual confounding from unmeasured factors, such as renal replacement therapy, cannot be excluded. Subsequent follow-up should focus on the progression of renal dysfunction and the occurrence of renal replacement therapy. Third, TyG index, 2D RV strain, and 3D volumetric indices were assessed only at baseline, long-term imaging and metabolic profiling may better capture dynamic changes and refine risk prediction. This limitation is particularly relevant for the TyG index, as a notable proportion of patients in our cohort, particularly those with CKD, were insulin-dependent. Exogenous insulin administration can induce glucose fluctuations, potentially influencing the visit-to-visit variability of the TyG index, which we were unable to assess. In future analyses, patients will be stratified according to different stages of CKD to further clarify the relationship between CKD severity and RV function. Fifth, certain inflammatory biomarkers (e.g., hsCRP, IL–6) were not available, which could have enriched the mechanistic interpretation of cardiometabolic pathways. Future studies should aim to validate these findings in larger, multicenter

cohorts. Additionally, prospective studies with long-term follow-up data are needed to assess the dynamic changes in the TyG index and RV function over time, and their impact on long-term outcomes in HFpEF patients, particularly those with CKD.

6 | Conclusions

In conclusion, combining the TyG index and 2D–RVFWLS may help identify higher-risk HFpEF phenotypes, especially in CKD, offering incremental prognostic value beyond either measure alone. We further need external validation and threshold optimization before clinical implementatin.

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Ethics Statement

This study complied with the Declaration of Helsinki and was approved by the Ethics Committee of Gansu Provincial Hospital (Approval No. 2025–566). Since all data were collected for clinical purposes and no invasive procedures were performed, no specific written consent form was necessary for this study.

Consent

All authors reviewed and approved the final version of this manuscript for publication.

Conflicts of Interest

The authors declare no conflicts of interests.

Data Availability Statement

Datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supplemental Figure S1. Cumulative incidence curves of heart failure rehospitalization by TyG index and RVFWLS (Fine–Gray models) Panel A: baseline TyG tertiles, Panel B: 3D RVEF dichotomized at the cohort median (high vs low), Panel C: Absolute RVFWLS dichotomized at the median (high vs low). Panel D: four groups by median-split TyG (high vs low) and RVFWLS (high vs low). Statistical differences between strata were tested with Gray's method (*p* values indicated in each panel). **Supplemental Figure S2**. Subgroup Sensitivity Analyses of TyG Index, RVFWLS, RVEF in HFpEF Forest plots showing HRs and 95 % CIs for the association of (A) TyG index per 1-unit increase, (B) Absolute 2D–RVFWLS (per 1% absolute increase), and (C) RVEF per 1% decrease with the composite endpoint of cardiovascular death or heart failure rehospitalization, across clinical subgroups. Subgroups include BMI (Normal vs. Overweight), Sex (Male vs. Female), and eGFR (≥ 60 vs. < 60 mL/min/1.73 m^2). Interaction *p* values are shown on the right of each panel. The vertical dashed line indicates HR $= 1$. Arrows denote point estimates with CIs extending beyond the plotting region.The Models are adjusted for age, sex, BMI, eGFR, log (NT-proBNP), HDL-C, RV S′, and Mitral E/A. **Supplemental Figure S3**. Bland–Altman plots for reproducibility of RVFWLS and RVEF (A–B) the inter- and intra-observer reproducibility of RVFWLS. (C–D), the inter- and intra-observer reproducibility of RVEF. The Bland–Altman plots depict the mean bias (blue dashed line) and the 95 % limits of agreement (red dashed lines, mean $\pm$ 1.96 SD). Each dot represents a single patient measureTable 1. Baseline characteristics of patients with HFpEF stratified by CKD status Supporting Information Supporting Information Supporting Information